

Optimization of an aeronautic propulsion system

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Content

- I. Context and system presentation
- II. Superconducting filter
- III. Optimization of the filter

I. Context and system presentation

MEA: More Electric Aircraft

Aims at replacing hydraulics, pneumatics or thermics motors and rotors by electric means.

Stakes:

- Lowering greenhouse gases emission
- Lowering operating and exploiting costs
- Safe and comfortable aircrafts

Issues:

- Stability issues



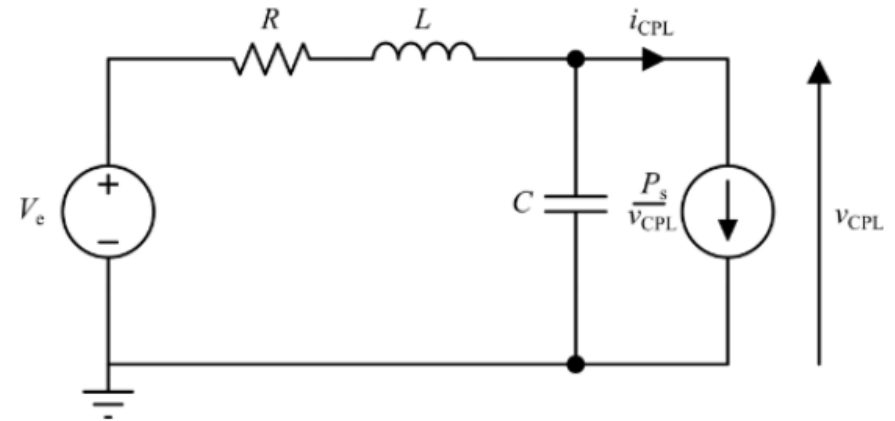
[2] A. Russo and A. Cavallo, "Supercapacitor stability and control for More Electric Aircraft application," *2020 European Control Conference (ECC)*, St. Petersburg, Russia, 2020, pp. 1909-1914, doi: 10.23919/ECC51009.2020.9143927

Toward an RLC system with a CPL

DC microgrids are commonly found in the power systems of aircrafts. They are controlled to maintain constant voltage, current and power and they can be simplified as constant power load (CPL).

Stakes:

- Stability of the system is unsure



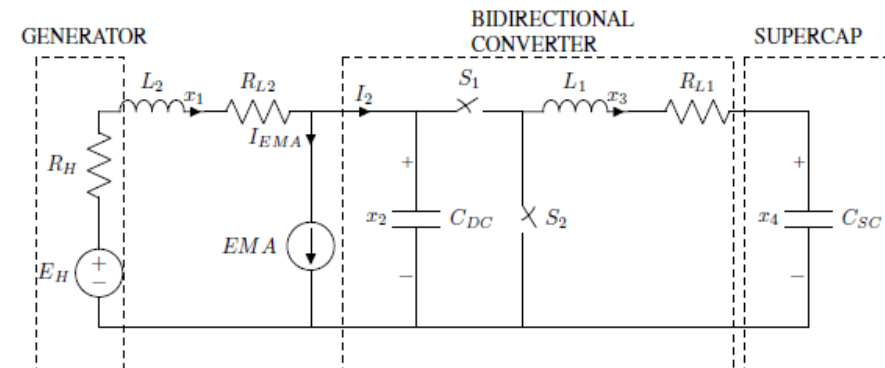
[1] L. Herrera, W. Zhang and J. Wang, "Stability Analysis and Controller Design of DC Microgrids With Constant Power Loads," in *IEEE Transactions on Smart Grid*, vol. 8, no. 2, pp. 881-888, March 2017, doi: 10.1109/TSG.2015.2457909.

[3] G. Ensermu and A. Bhattacharya, "Control and Stability Analysis of DC Micro-grids with Constant Power Loads and Source Disturbances," *2018 IEEE International Conference on Power Electronics, Drives and Energy Systems (PEDES)*, Chennai, India, 2018, pp. 1-6, doi: 10.1109/PEDES.2018.8707605.

Passive solution

Supercapacitor stability for more electric aircraft application

- Problem: Some instability can occur
- Solution: Using a **bidirectional converter** to regulate the power flow between the power generator and a supercapacitor.
- Features
 - soothing action on generator current
 - Stability of the open-loop



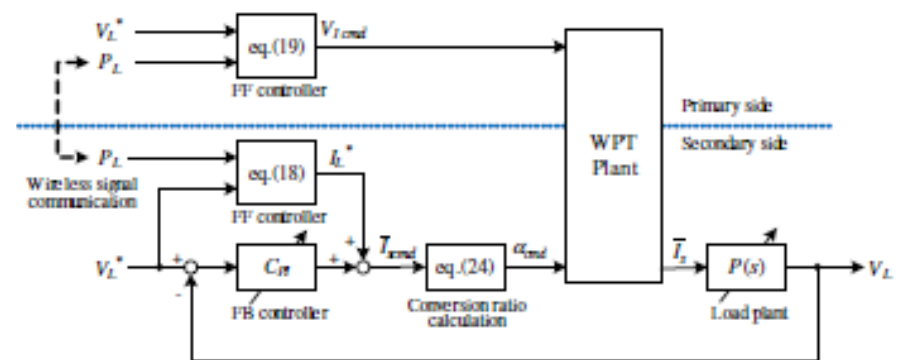
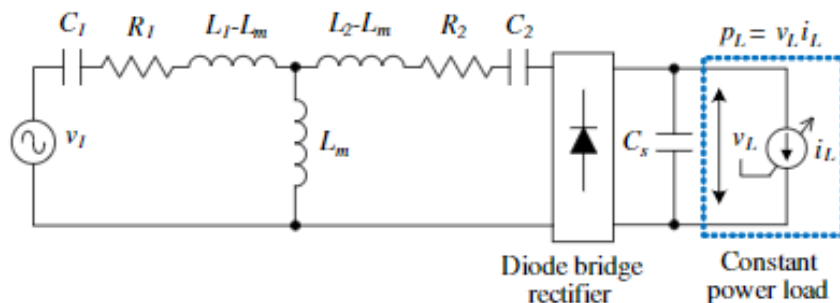
Bidirectional converter with supercapacitor schematic

[2] A. Russo and A. Cavallo, "Supercapacitor stability and control for More Electric Aircraft application," *2020 European Control Conference (ECC)*, St. Petersburg, Russia, 2020, pp. 1909-1914, doi: 10.23919/ECC51009.2020.9143927

Active solution

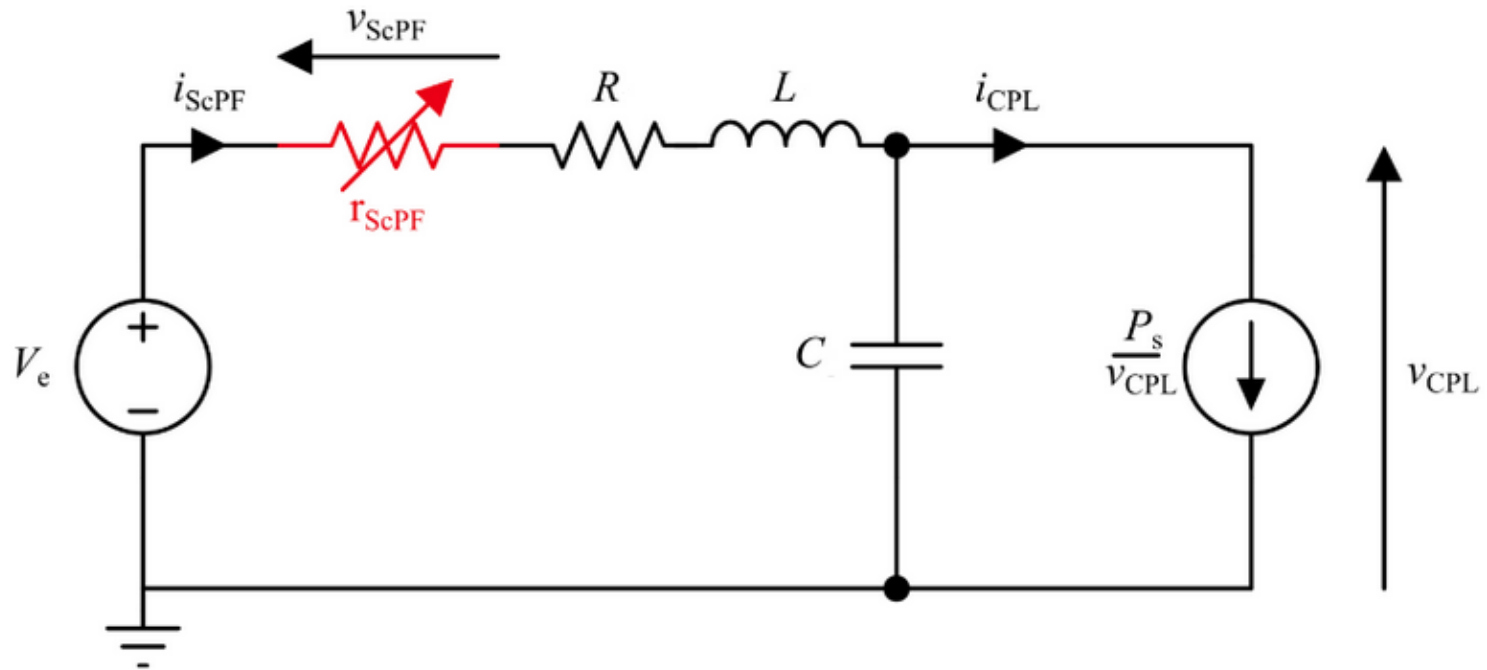
Load voltage control for wireless in-wheel motor

- Problem: The system without control is unstable
- Solution: Controller method
- Features: use 2 types of operation modes of the secondary converter (in the wheel part): diode bridge rectifier, to control the transmitting power + feedforward controller + 2 freedom controllers



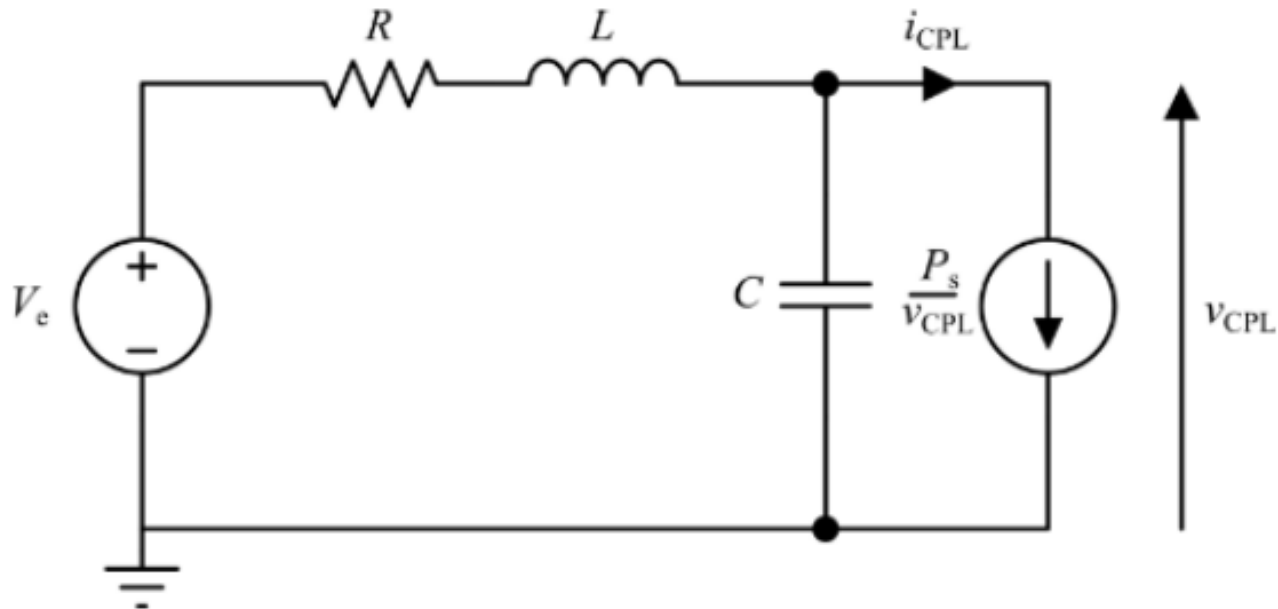
Other solution

Add a superconducting filter in the circuit to avoid instabilities



II. Superconducting filter

Stability condition for a DC circuit under constant power load

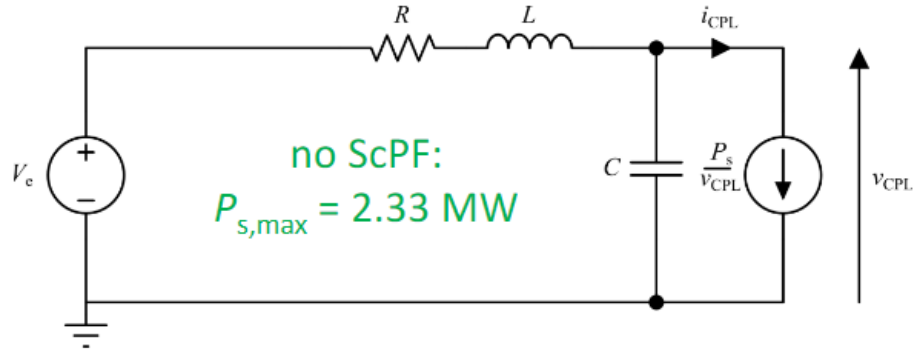


Stable if

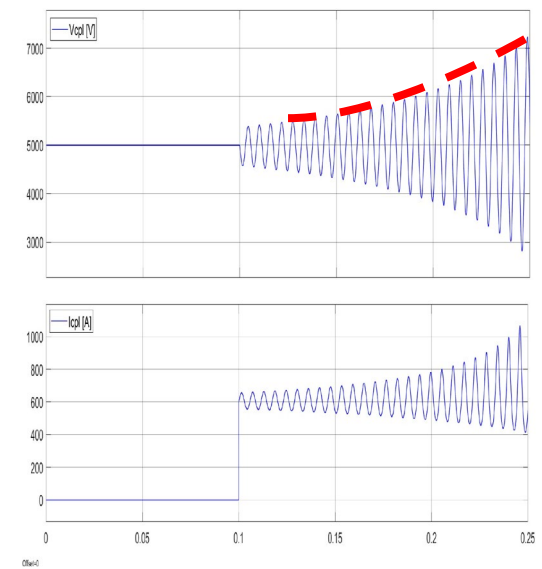
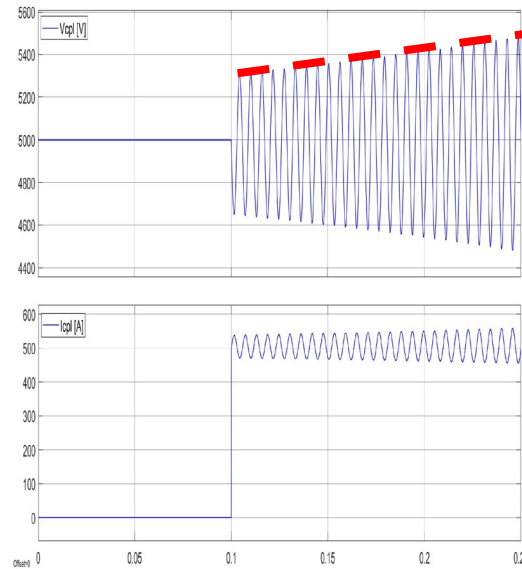
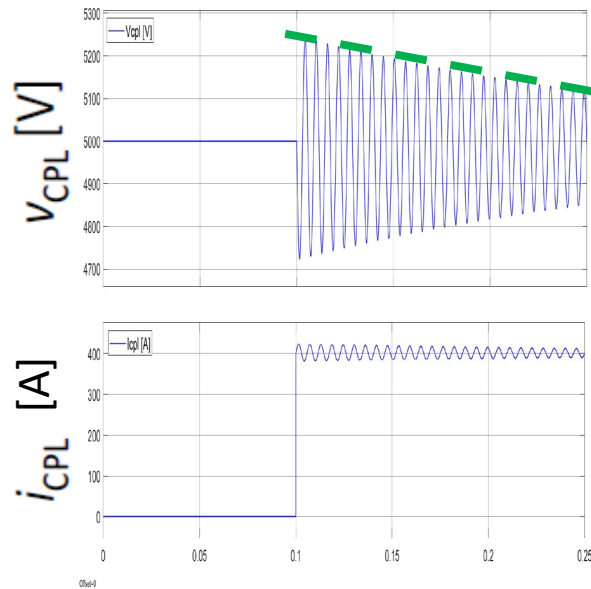
$$P_s \leq \frac{RC}{L} V_e^2$$

Numerical stability

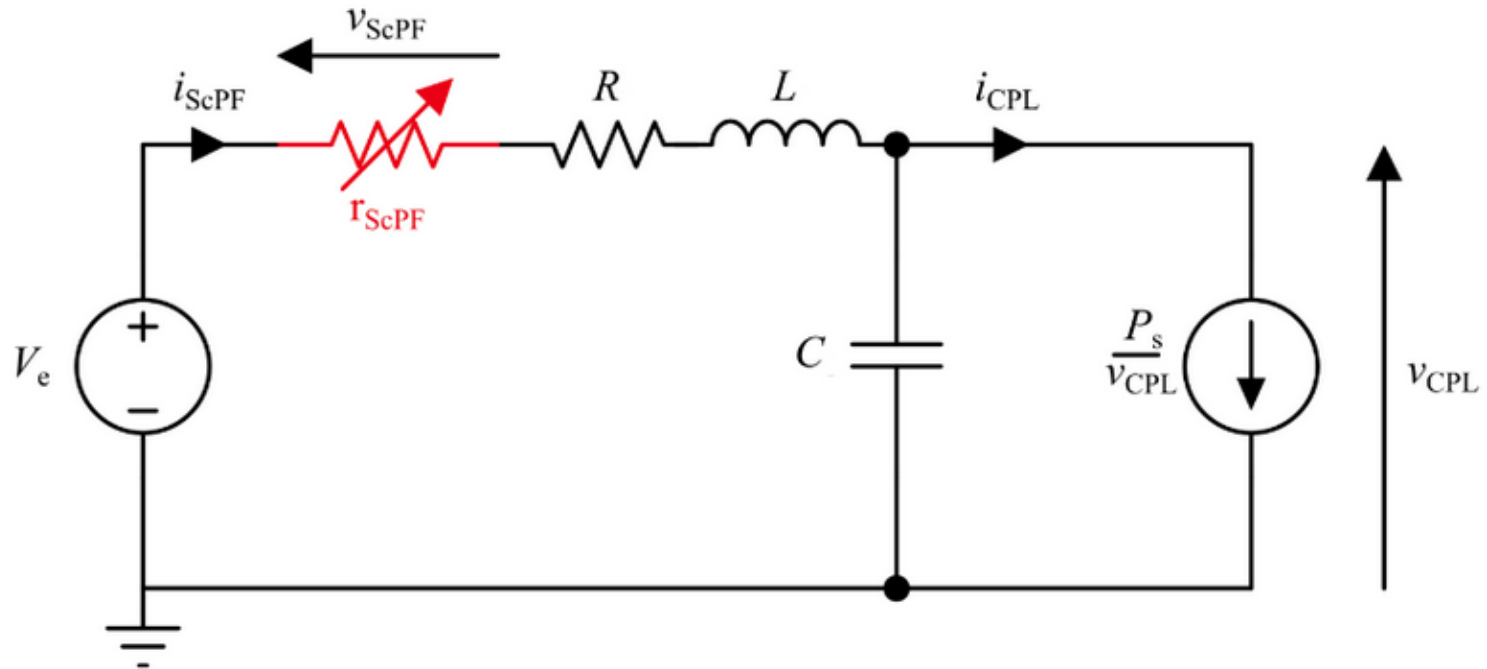
$$P_{s,max} = \frac{RC}{L} V_e^2$$



$P_s = 2 \text{ MW}$: stable $P_s = 2.5 \text{ MW}$: Unstable $P_s = 3 \text{ MW}$: Unstable



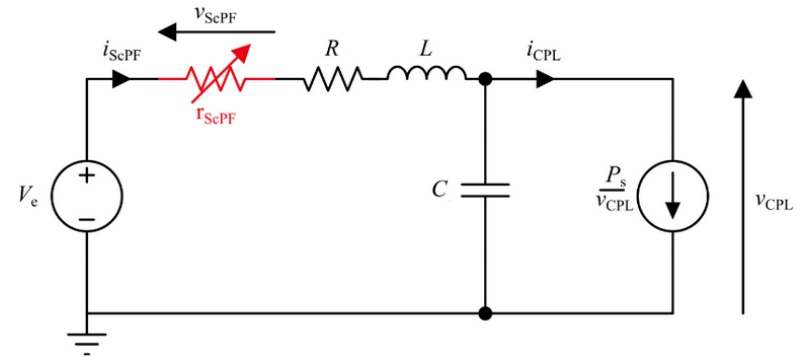
Stability condition for a DC circuit under constant power load with superconductor



Stable if:
$$P_s \leq \frac{(R + r_{supra}(I))C}{L} V_e^2$$

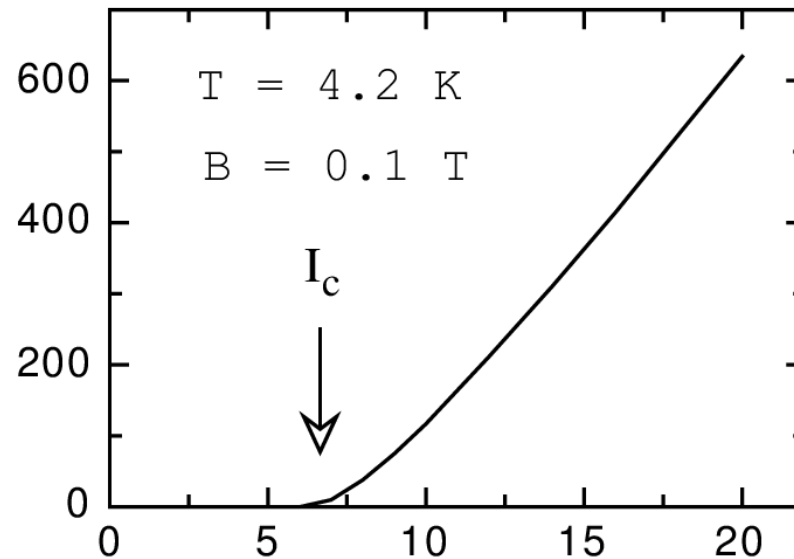
Advantages of the superconductor

$$P_s \leq \frac{(R + r_{supra}(I))C}{L} V_e^2$$



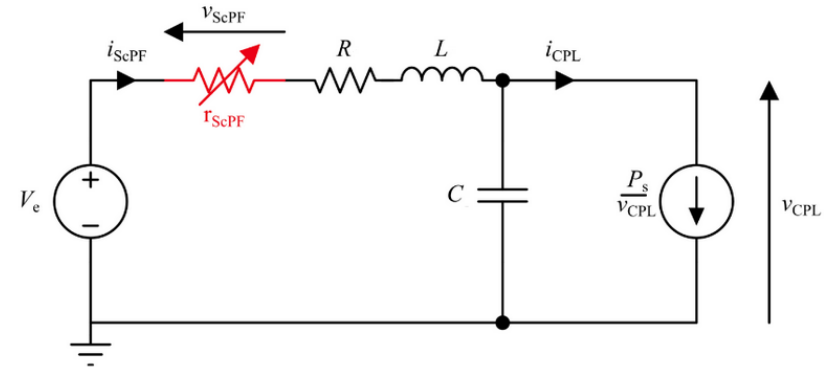
Characteristic UI of the superconductor:

⇒ Become resistant at a certain current

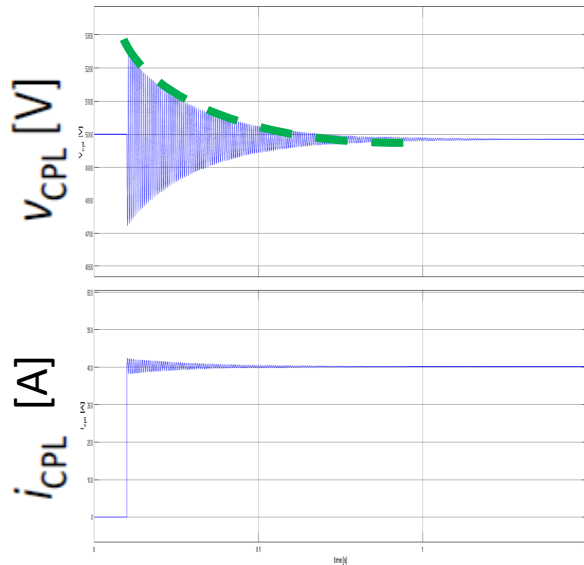


Numerical stability

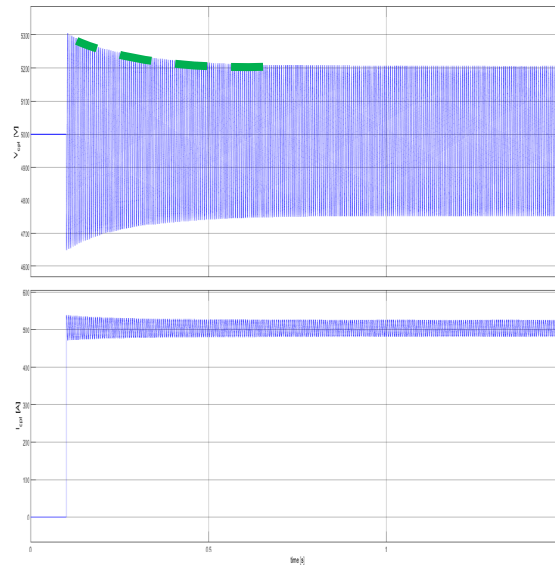
$$P_{s,max} = \frac{(R + r_{supra}(I))C}{L} V_e^2$$



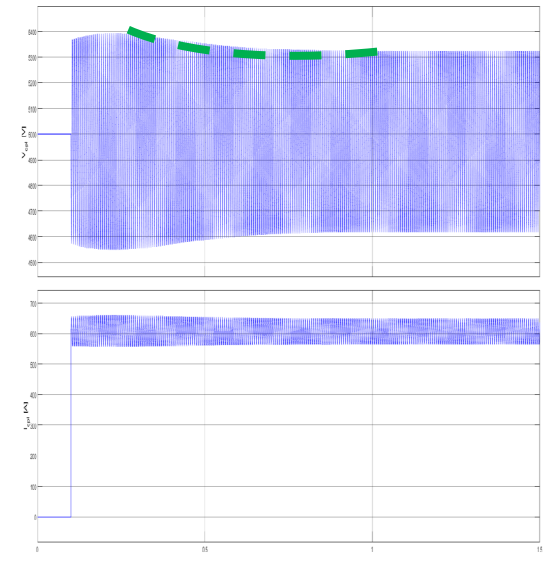
$P_s = 2 \text{ MW}$: stable



$P_s = 2.5 \text{ MW}$: Stable



$P_s = 3 \text{ MW}$: Stable



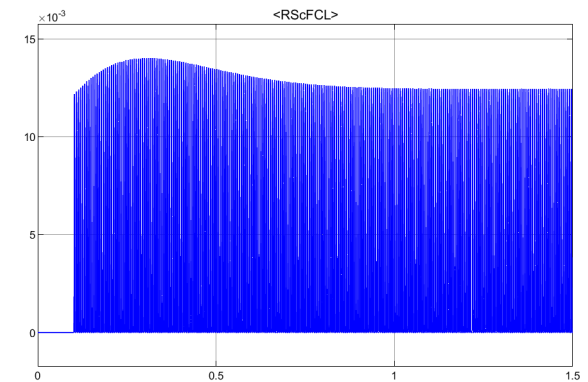
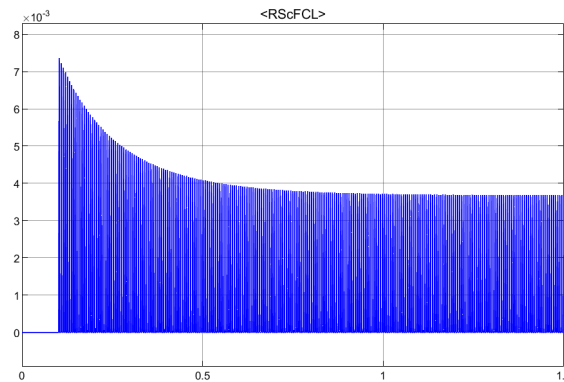
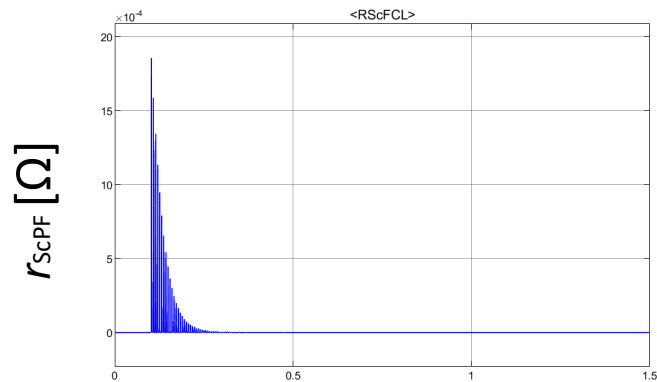
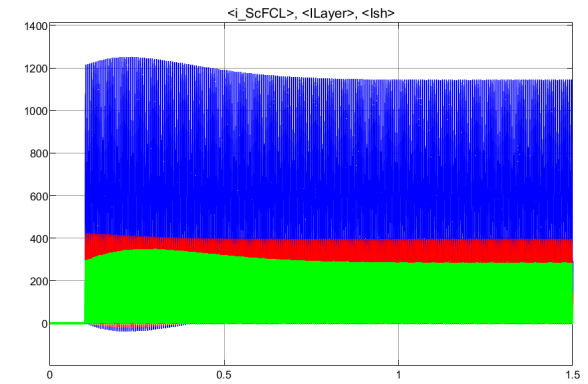
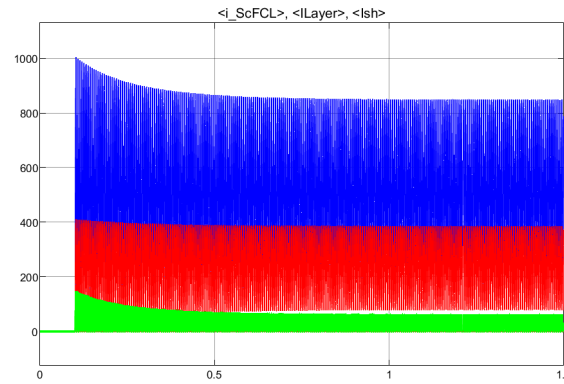
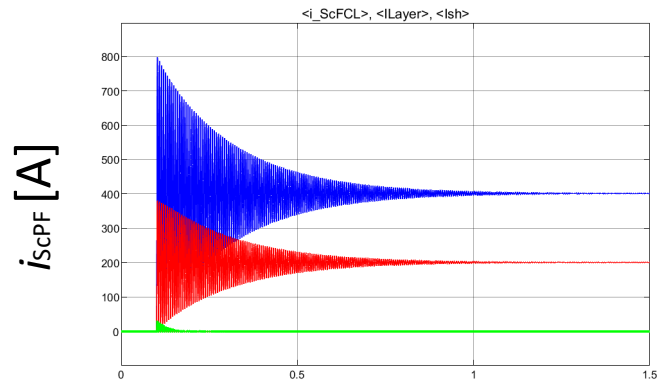
=> Stability for all power

ScPF behavior (simulated with Matlab)

$P_S = 2 \text{ MW}$: stable

$P_s = 2.5 \text{ MW}$: stable

$P_S = 3 \text{ MW}$: stable



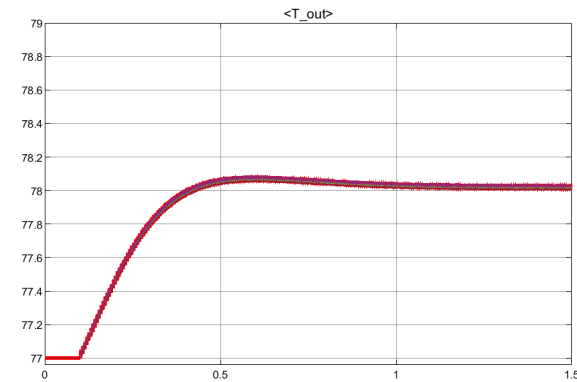
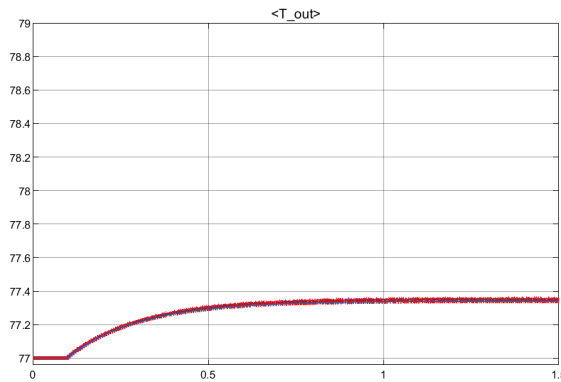
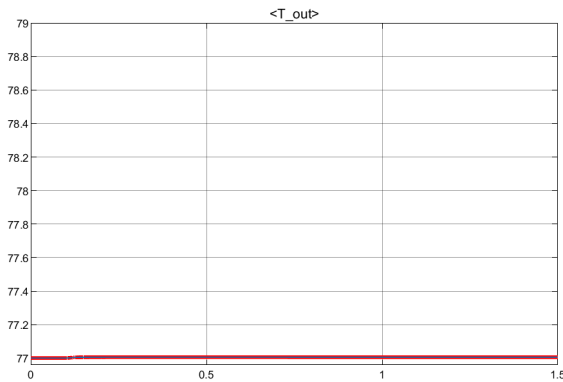
ScPF behavior (simulated with Matlab)

$P_s = 2 \text{ MW}$: **stable**

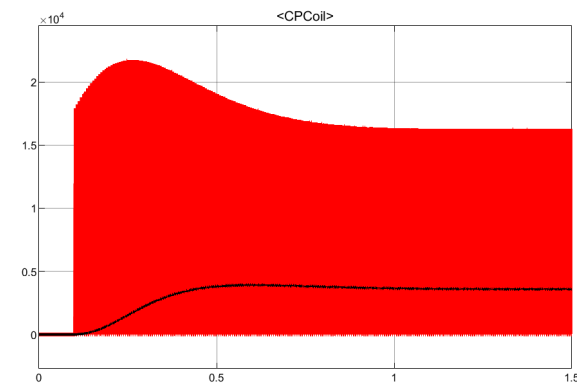
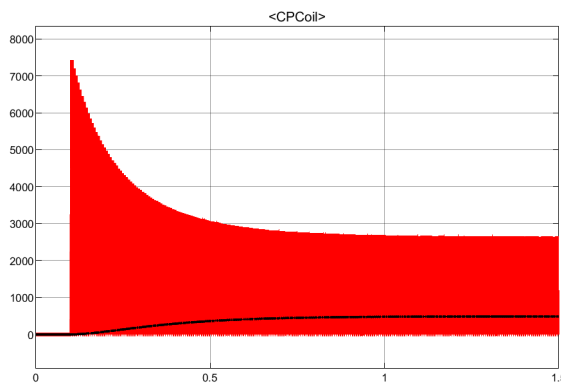
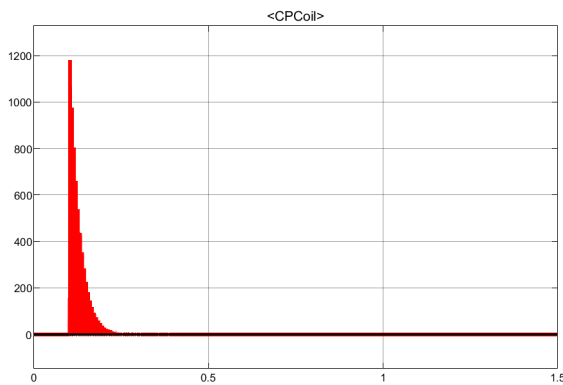
$P_s = 2.5 \text{ MW}$: **stable**

$P_s = 3 \text{ MW}$: **stable**

$T_{sc} \text{ [K]}$

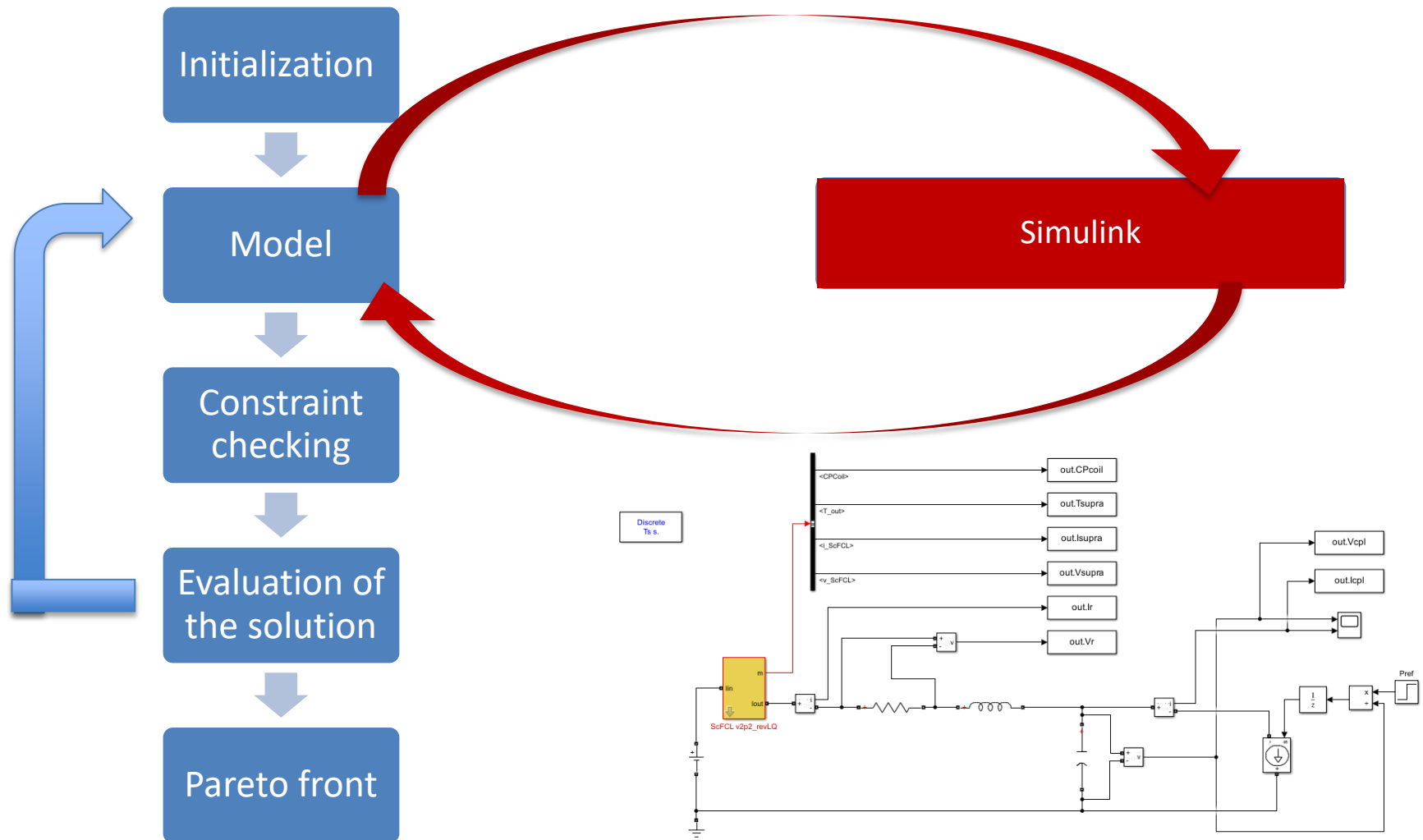


$Power_{super}, CP_{coil} \text{ [W]}$



III. Filter's optimization

Coupling between MOPSO and Simulink



Optimization according to R and -L

Goal: check the coupling

- No concrete physical interpretation

Variables

$$R \in [30 \cdot 10^{-3}; 50 \cdot 10^{-3}][\Omega]$$

$$L \in [0.2 \cdot 10^{-3}; 0.8 \cdot 10^{-3}][H]$$

Objectives

$$\min \begin{cases} R \\ -L \end{cases}$$

Constraints

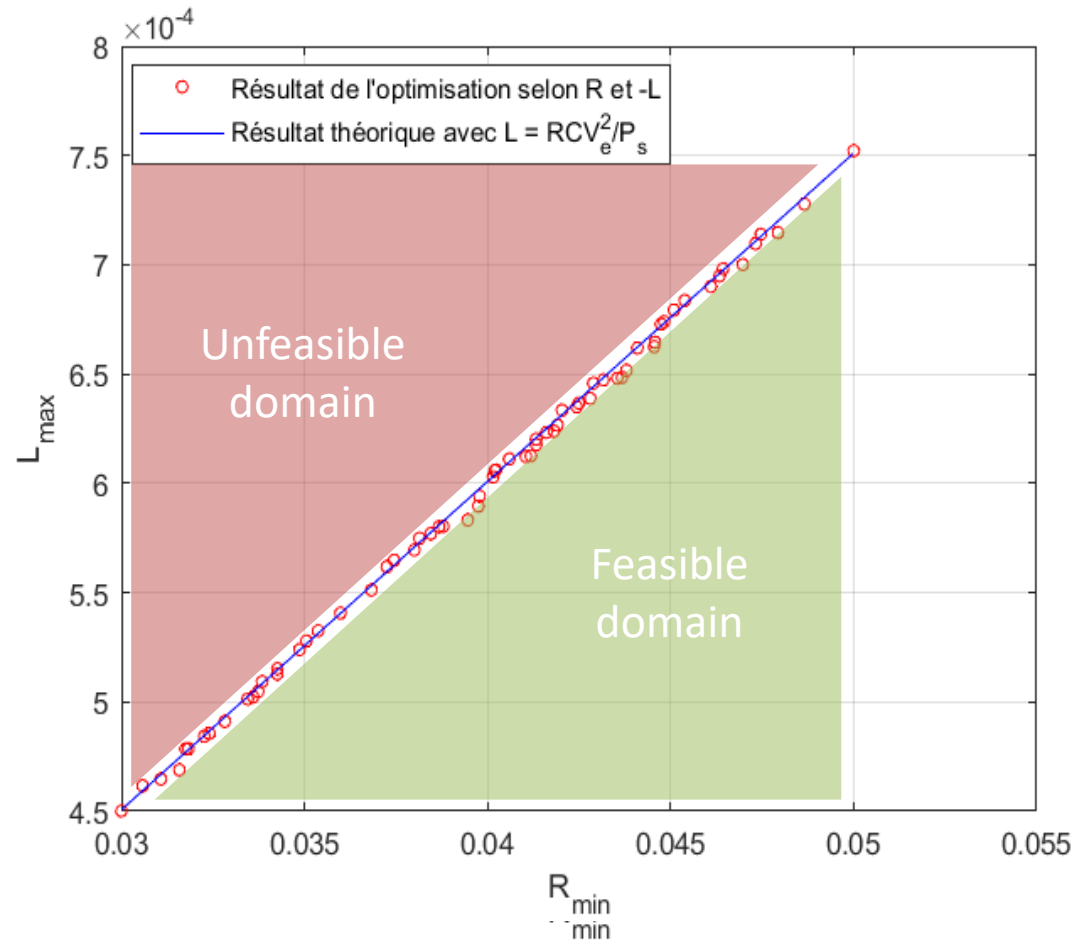
- Stability for $P = 2.33$ MW

Parameters

30 particles, 100 iterations

Optimization according to R et -L

$$P_s \leq \frac{RC}{L} V_e^2$$

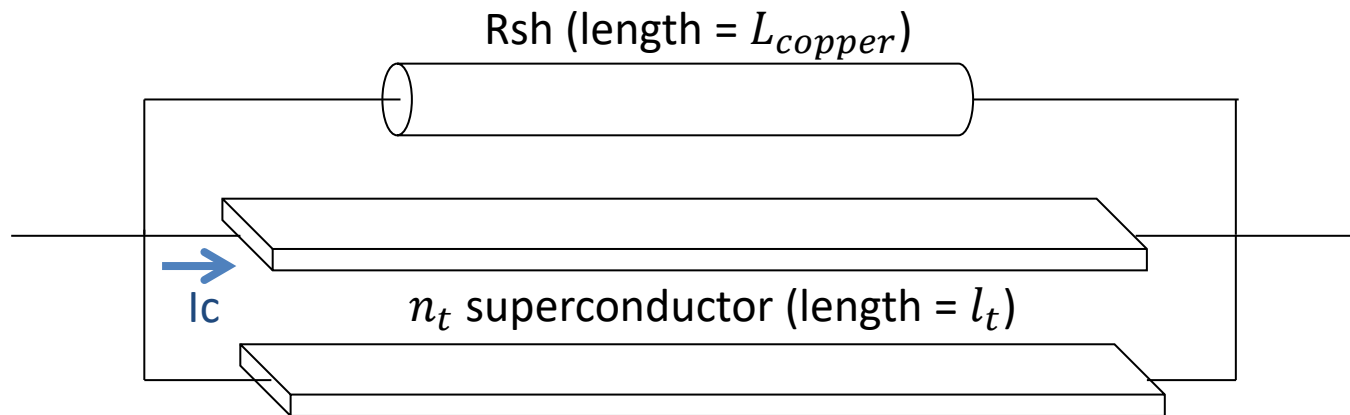


=> Coupling between Matlab and Simulink : Ok for the test

III. 1. Optimization Cost_copper / Cost_super

Cost_copper VS Cost_super

- Goal: Compromise between the cost of the material
- $P_{sc} = 100 \cdot P_{Cu}$ with P_{sc} price per meter of the superconductor and P_{Cu} price per meter of the copper
- Optimization : $C_{sc} = P_{sc} \cdot l_t \cdot n_t$ and $C_{cu} = L_{copper} \cdot P_{Cu}$



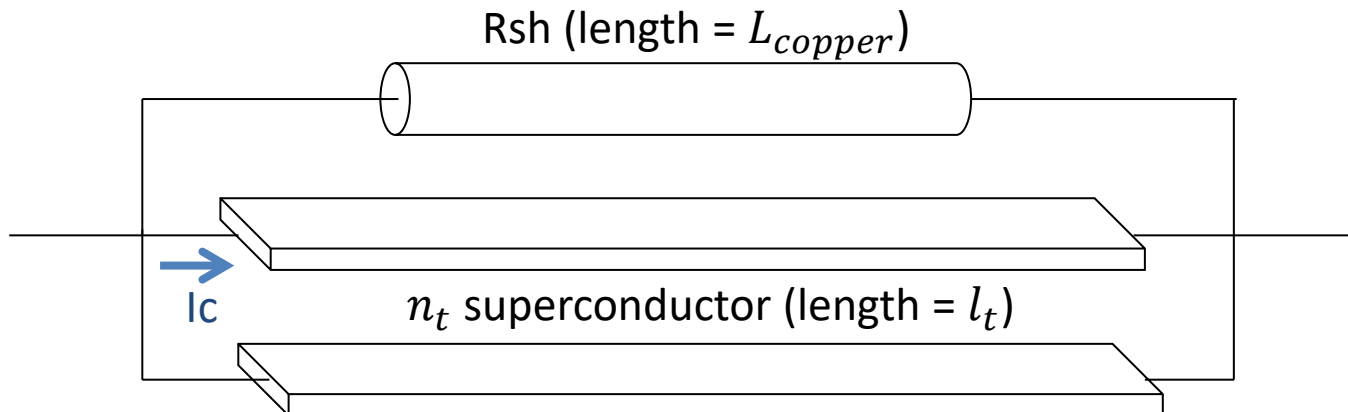
L_copper

- $L_{copper} = \frac{R_{sh}S}{\rho}$

with S the section of the wire and ρ the resistivity of the copper

- $S = \frac{n_t I_c}{J}$

with $J = 2 \text{ A/mm}^2$ and $I_c = 300 \text{ A}$ (maximum current into the superconductor)



Optimization according to C_{cu} and C_{sc}

Variables

$$n_t \in \llbracket 1; 3 \rrbracket$$

$$L_t \in [1; 100][\text{m}]$$

$$R_{sh} \in [0.1 \cdot 10^{-3}; 1][\Omega]$$

Objectives

$$\min \begin{cases} C_{sc} \\ C_{cu} \end{cases}$$

Constraints

- Stability for $P = 1.2 \cdot 170.5 \text{ kW}$
- $0 \leq V_{CPL}(t) \leq V_{max}$
- $I_{CPL}(t) \leq I_{max}$
- $T_{supra}(t) \leq T_{max}$

Parameters

100 particles, 100 iterations

$$V_{max} = 650 \text{ V}$$

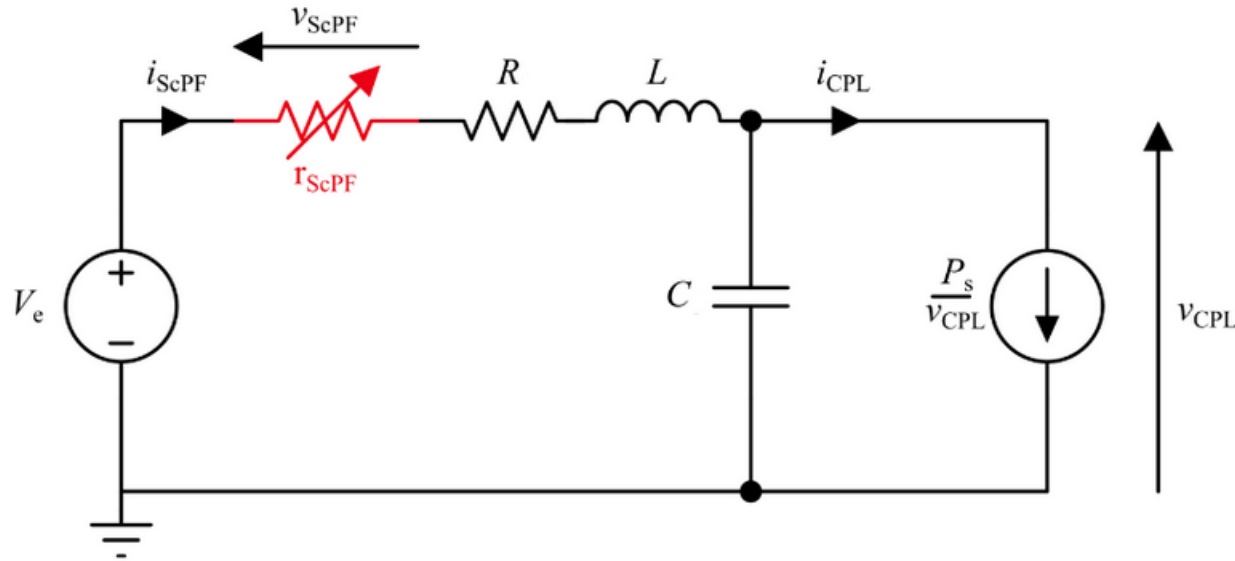
$$I_{max} = 630 \text{ A}$$

$$T_{max} = 90 \text{ K}$$

Parameters

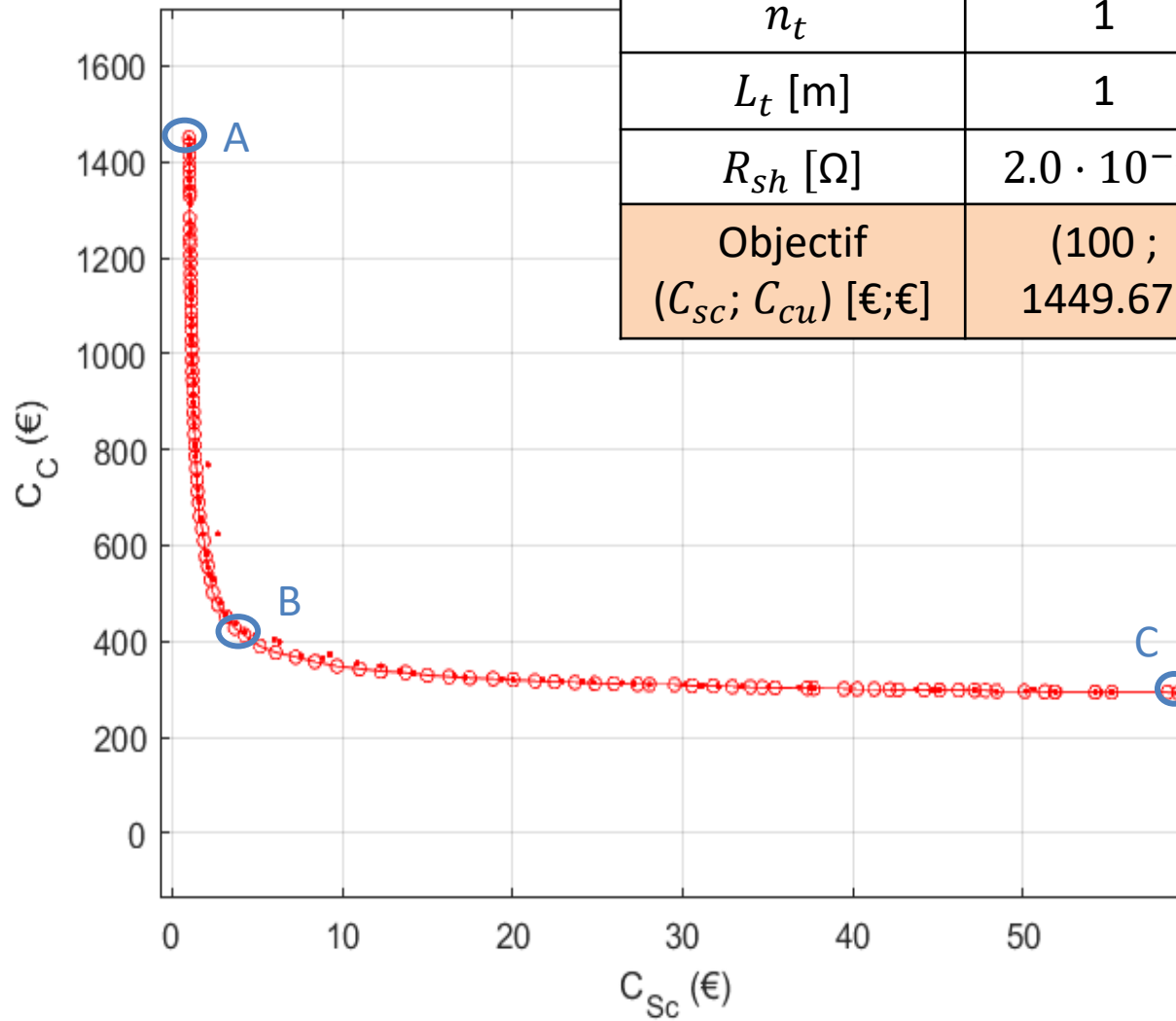
$$0 \leq V_{CPL}(t) \leq V_{max}$$

$$I_{CPL}(t) \leq I_{max}$$



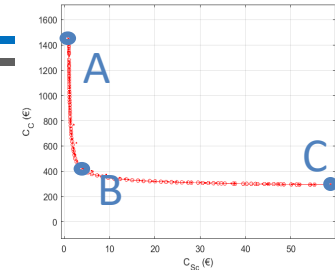
Name (unit)	$V_{max}(\text{V})$	$I_{max}(\text{A})$	$P_{Sref}(\text{W})$
Formula	$120\% V_{nom}$ $= 120\% \cdot 540$	$200\% I_{nom}$ $= 2 \frac{P_{Sref}}{V_{nom}}$	$\frac{RC}{L} V_e^2$
Value	650	630	170 500

$C_{sc} - C_{cu}$

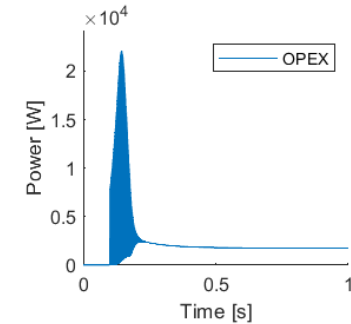
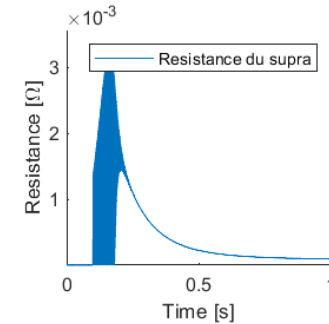
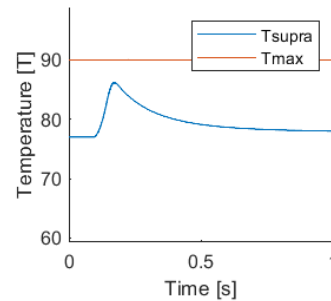
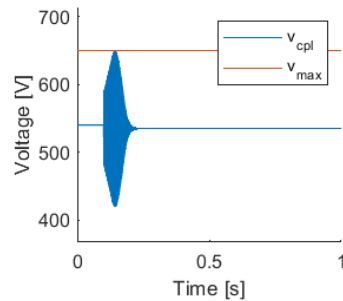
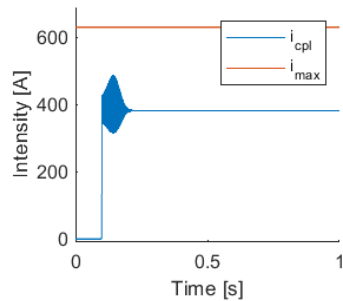


	A	B	C
n_t	1	1	1
L_t [m]	1	4.24	59.12
R_{sh} [Ω]	$2.0 \cdot 10^{-2}$	$5.7 \cdot 10^{-3}$	$4.0 \cdot 10^{-3}$
Objectif (C_{sc} ; C_{cu}) [€;€]	(100 ; 1449.67)	(423.8 ; 418.86)	(5912.12 ; 291.907)

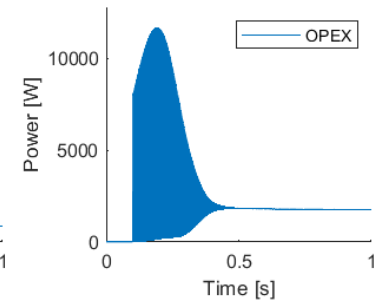
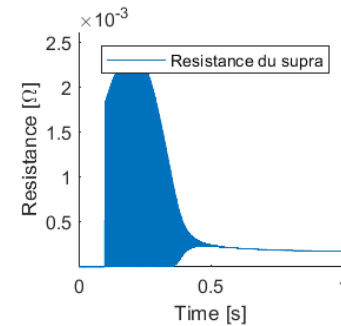
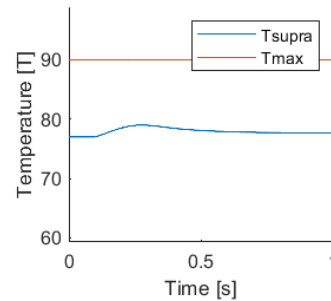
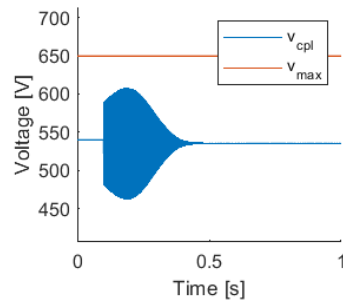
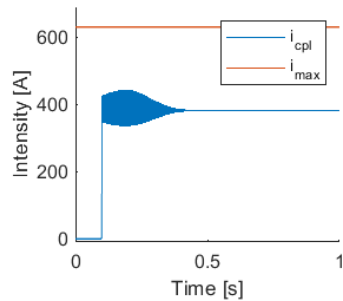
Particular points



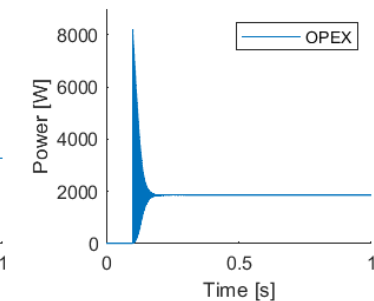
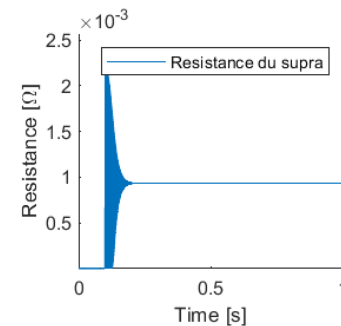
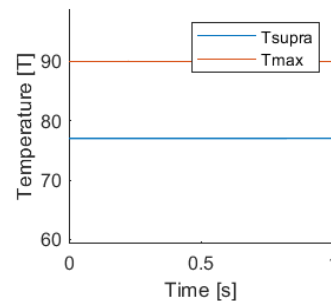
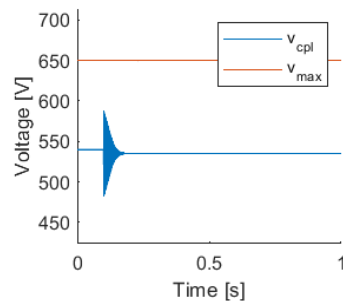
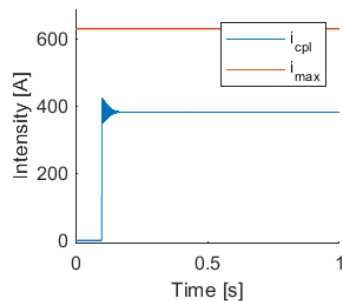
Point A



Point B



Point C



III. 2. Optimization CAPEX / Power_{CPL}

Interests to increase the power of the CPL

- A break of one reactor: how many power can we have with another one?
 - Pressurisation problem, climatisation
 - During the landing and the take-off: more power needed in the plane
- => How the CAPEX influence the overpower allowed?

CAPEX

CAPEX = Capital expenditure

Formula

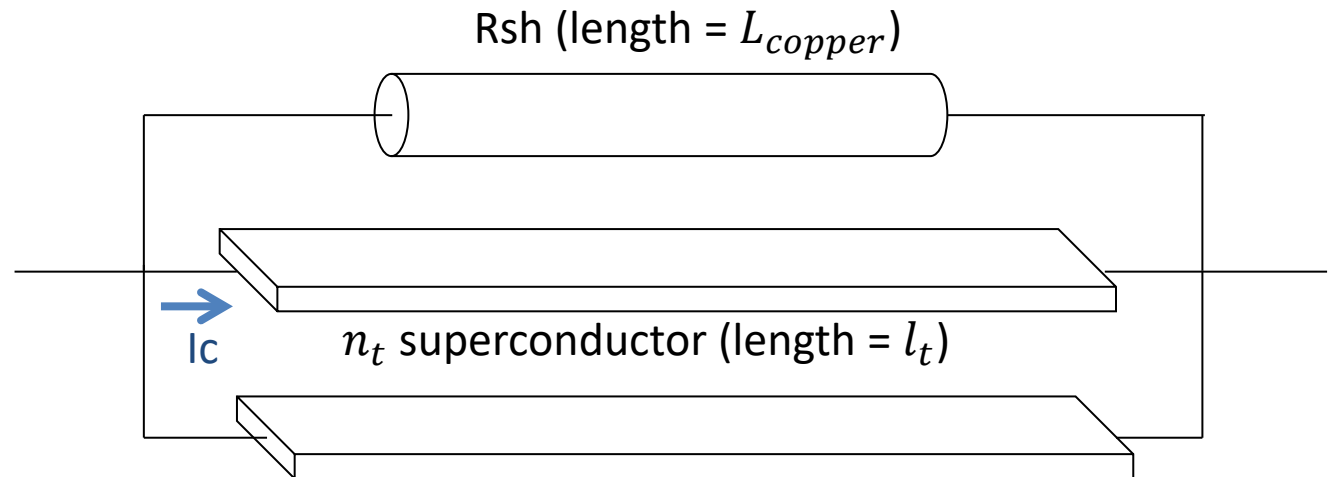
$$CAPEX = C_{sc} + C_{cu}$$

Hypothesis

- $P_{sc} = 100 \cdot P_c$

- $L_{copper} = \frac{R_{sh}S}{\rho}$

- $S = \frac{n_t I_c}{J}$



Optimization according to CAPEX and power of the constant power load

Variables

$$n_t \in \llbracket 1; 3 \rrbracket$$

$$L_t \in [1; 100][\text{m}]$$

$$R_{sh} \in [0.1 \cdot 10^{-3}; 1][\Omega]$$

$$\alpha \in [1; 2.5]$$

Objectives

$$\min \begin{cases} -\alpha \\ CAPEX \end{cases}$$

Constraints

- Stability for $P_{CPL} = \alpha \cdot 170.5 \text{ kW}$
- $0 \leq V_{CPL}(t) \leq V_{max}$
- $I_{CPL}(t) \leq I_{max}$
- $T_{supra}(t) \leq T_{max}$

Parameters

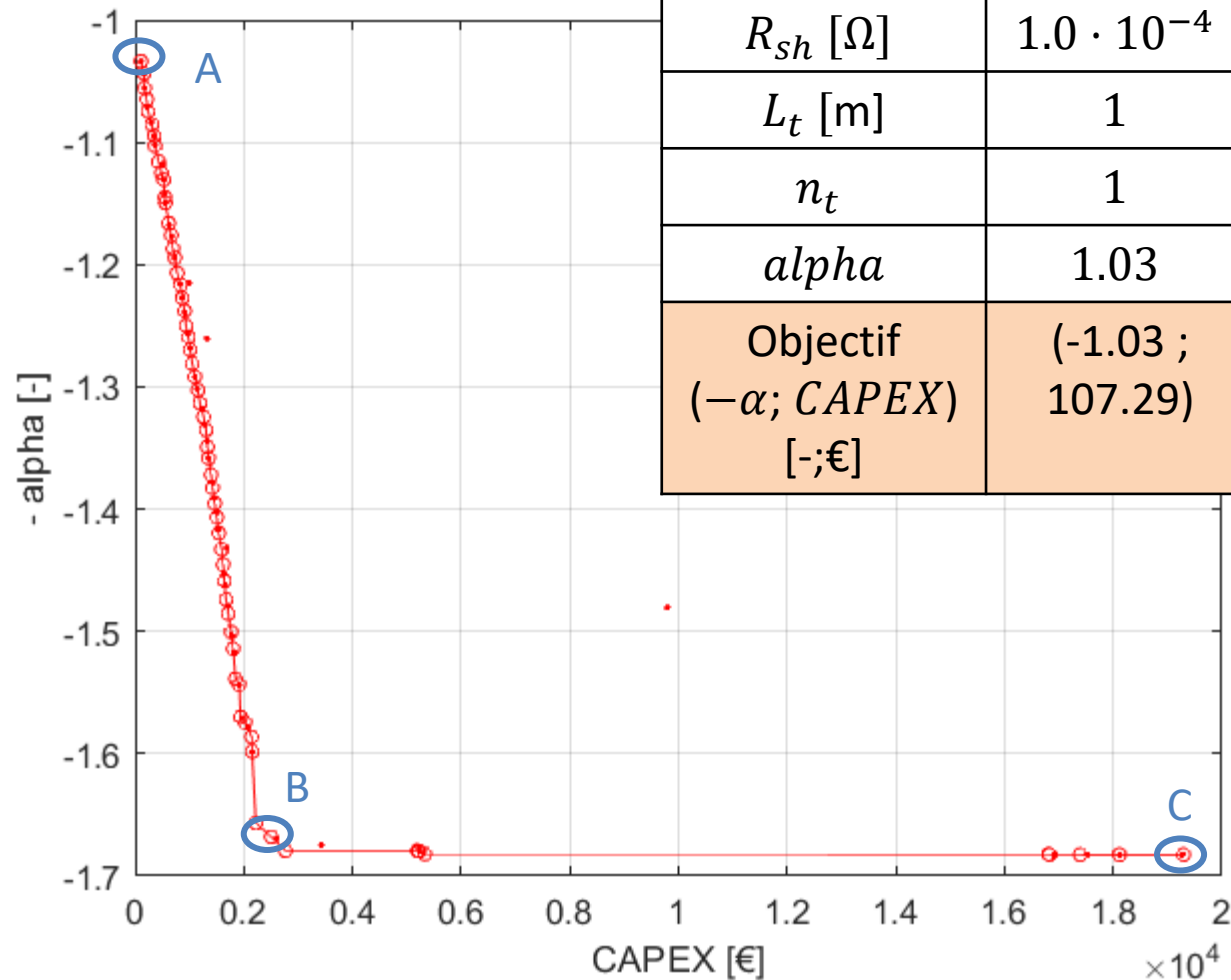
50 particules, 200 iterations

$$V_{max} = 650 \text{ V}$$

$$I_{max} = 630 \text{ A}$$

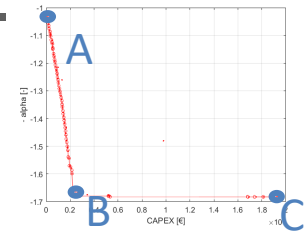
$$T_{max} = 90 \text{ K}$$

CAPEX — — α

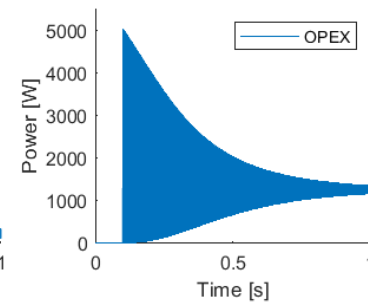
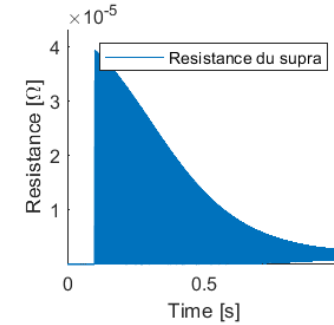
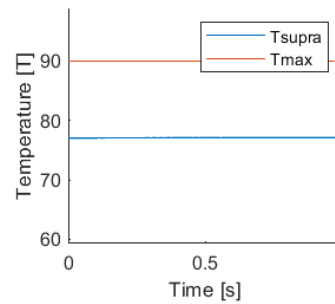
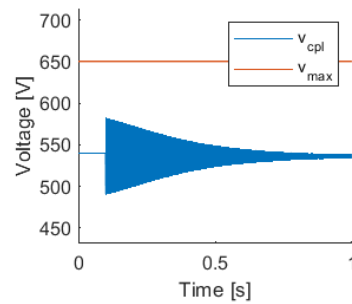
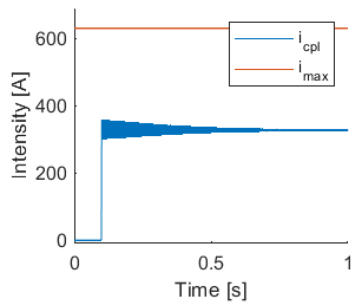


	A	B	C
$R_{sh} [\Omega]$	$1.0 \cdot 10^{-4}$	$2.0 \cdot 10^{-2}$	$1.1 \cdot 10^{-1}$
$L_t [m]$	1	10.3	56.2
n_t	1	1	2
α	1.03	1.67	1.68
Objectif ($-\alpha$; CAPEX) [-;€]	(-1.03 ; 107.29)	(-1.67 ; 2502.45)	(-1.68 ; 19276)

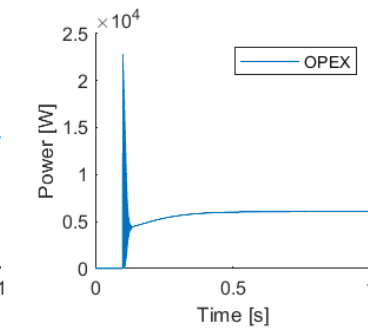
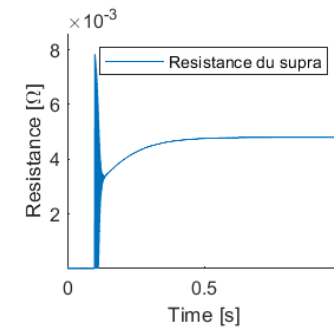
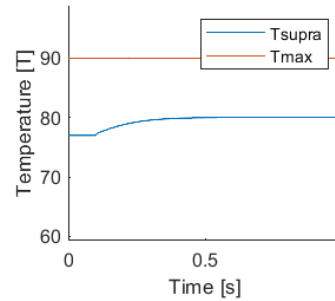
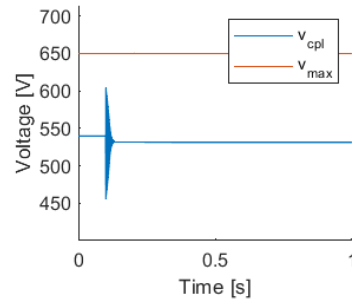
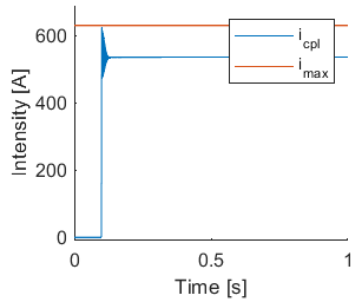
Particular points



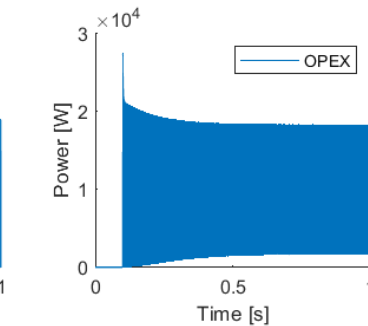
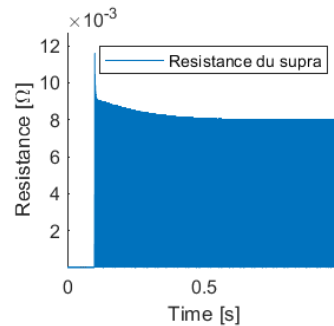
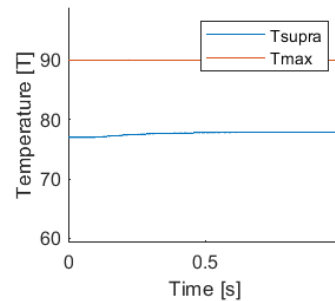
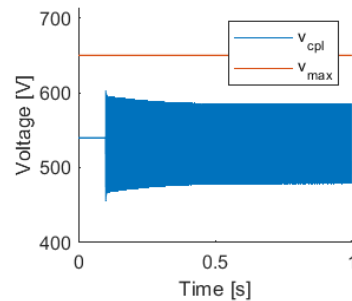
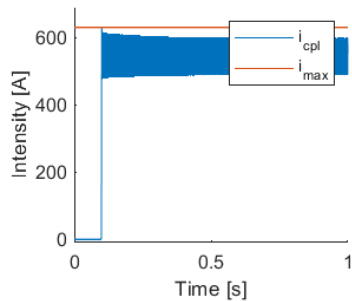
Point A



Point B



Point C



III. 3. Optimization CAPEX / OPEX

Life-cycle approach

- *CAPEX* = Capital cost
- *OPEX* = Operation and maintenance cost
- **Goal:** Have an overview of the cost of the system during its full life-cycle and find a compromise between the cost of maintenance and the cost of investment



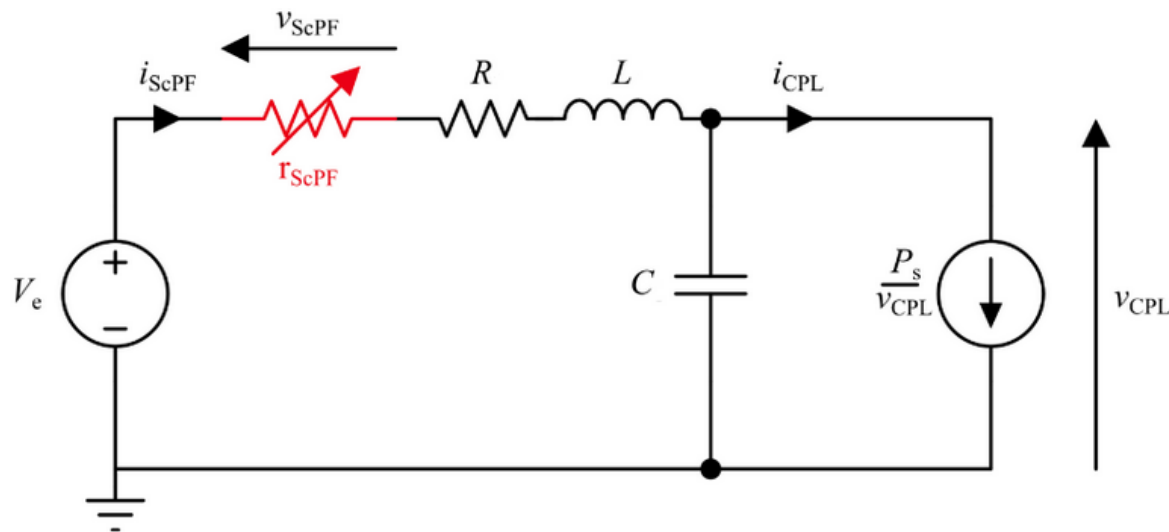
OPEX

Formula

$$OPEX = CP_{coil} + V_{ScPF} \cdot I_{ScPF} + V_R \cdot I_R$$

Hypothesis

- Mean value of the sum
- CPcoil: power use by the cooling system



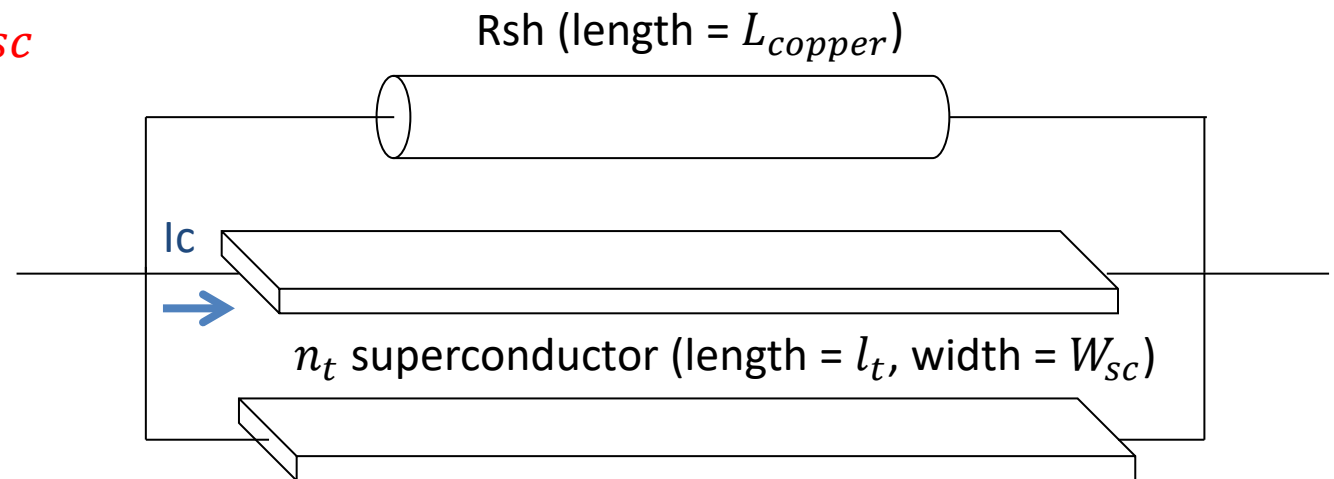
CAPEX

Formula

$$CAPEX = C_{sc} + C_{cu}$$

Hypothesis

- $P_{sc} = 100 \cdot P_c$
- $C_{sc} = l_t n_t P_{sc} W_{sc}$



Optimization according to CAPEX and OPEX

Variables

$$n_t \in \llbracket 1; 3 \rrbracket$$

$$L_t \in [1; 100][\text{m}]$$

$$R_{sh} \in [0.1 \cdot 10^{-3}; 1][\Omega]$$

Superconductor type :

I_c	75	112.5	150	225	300	450
W_{sc}	2	3	4	6	12	12

Objectives

$$\min \begin{cases} OPEX \\ CAPEX \end{cases}$$

Constraints

- Stability for $P = 1.2 \cdot 170.5 \text{ kW}$
- $0 \leq V_{CPL}(t) \leq V_{max}$
- $I_{CPL}(t) \leq I_{max}$
- $T_{supra}(t) \leq T_{max}$

Parameters

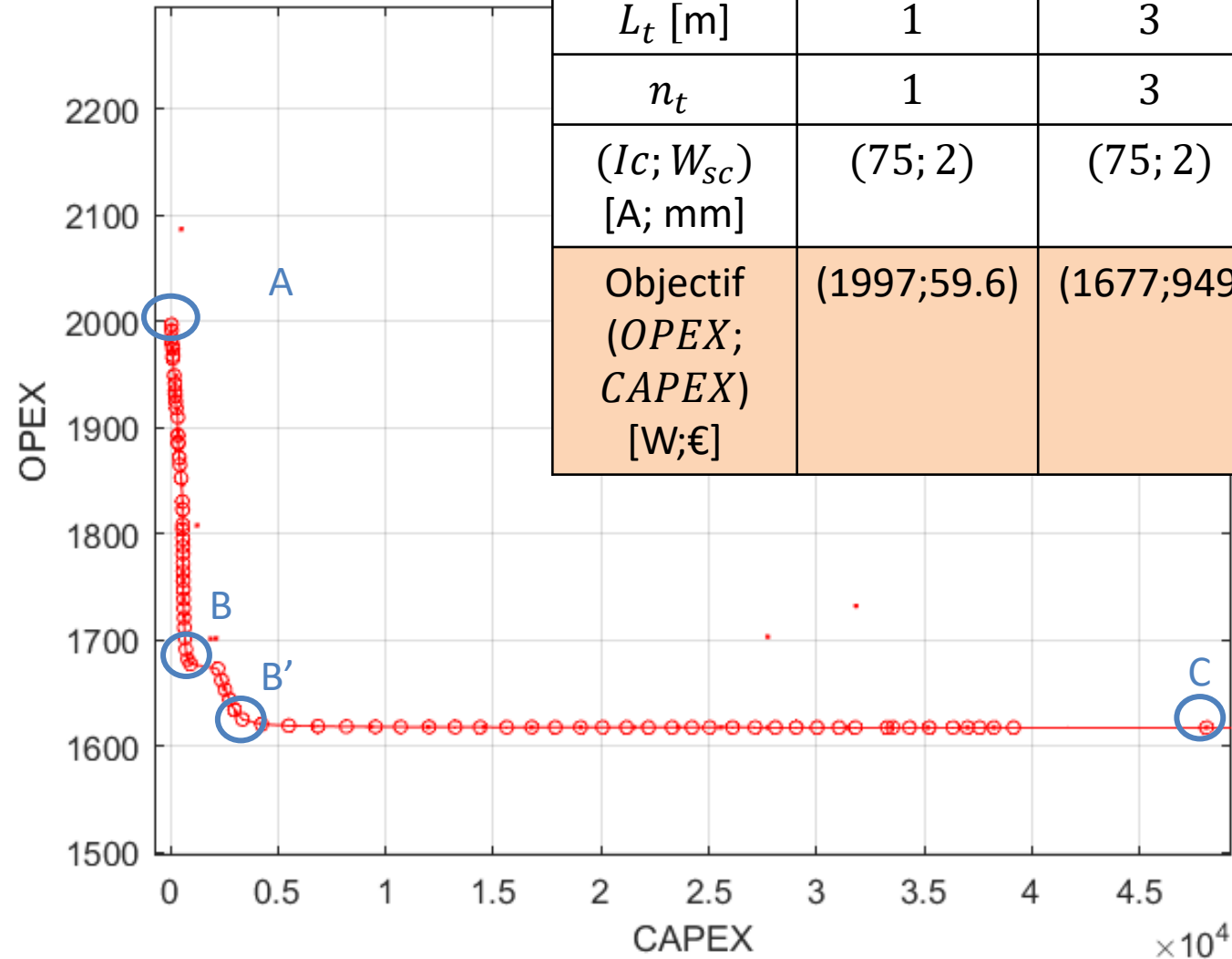
40 particles, 180 iterations

$$V_{max} = 650 \text{ V}$$

$$I_{max} = 630 \text{ A}$$

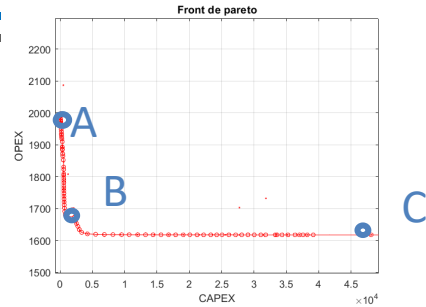
$$T_{max} = 90 \text{ K}$$

CAPEX - OPEX

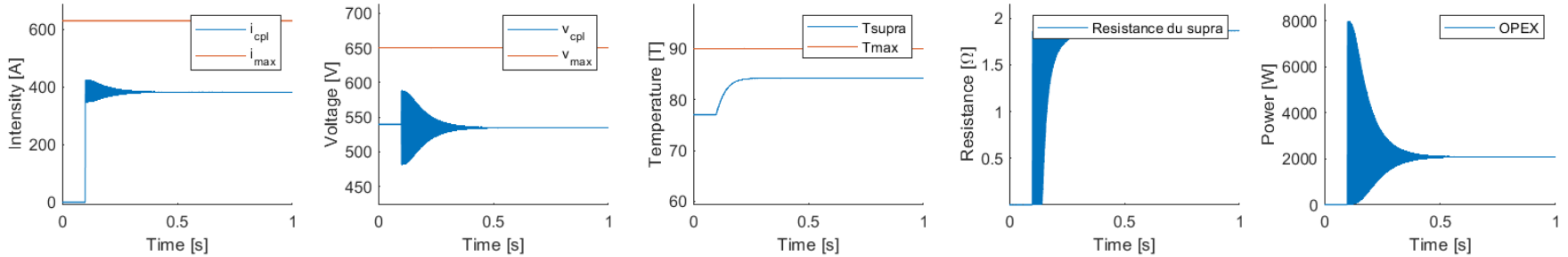


	A	B	B'	C
$R_{sh} [\Omega]$	$2.3 \cdot 10^{-3}$	$1.0 \cdot 10^{-2}$	$1.6 \cdot 10^{-2}$	$5.9 \cdot 10^{-1}$
$L_t [m]$	1	3	26	22.3
n_t	1	3	3	3
$(I_c; W_{sc})$ [A; mm]	(75; 2)	(75; 2)	(112.5; 3)	(112.5; 3)
Objectif (OPEX; CAPEX) [W;€]	(1997;59.6)	(1677;949)	(1625; 3349)	(1617 ; 49905)

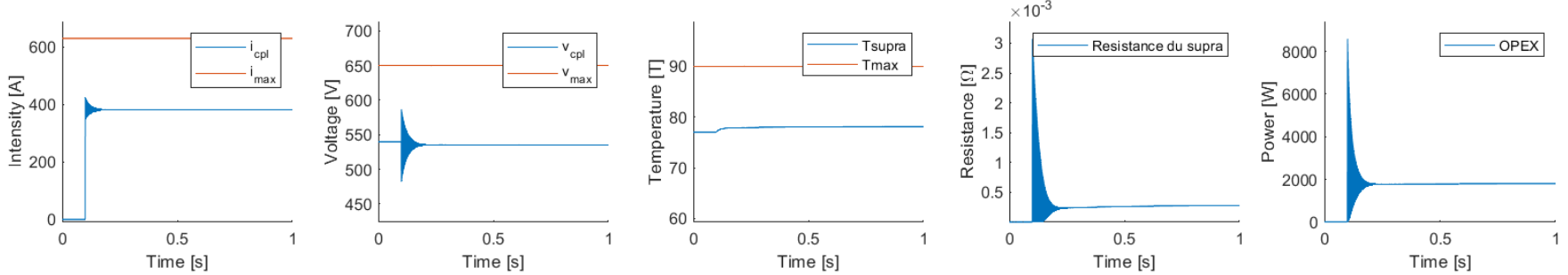
Particular points



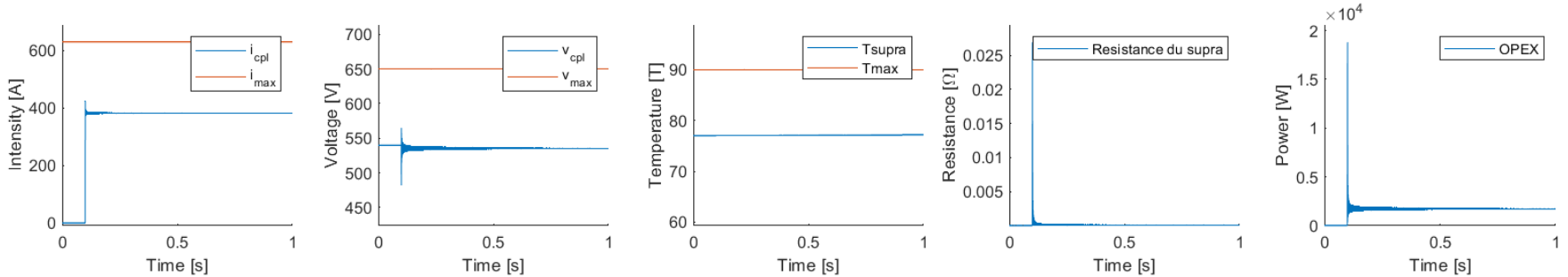
Point A



Point B



Point C

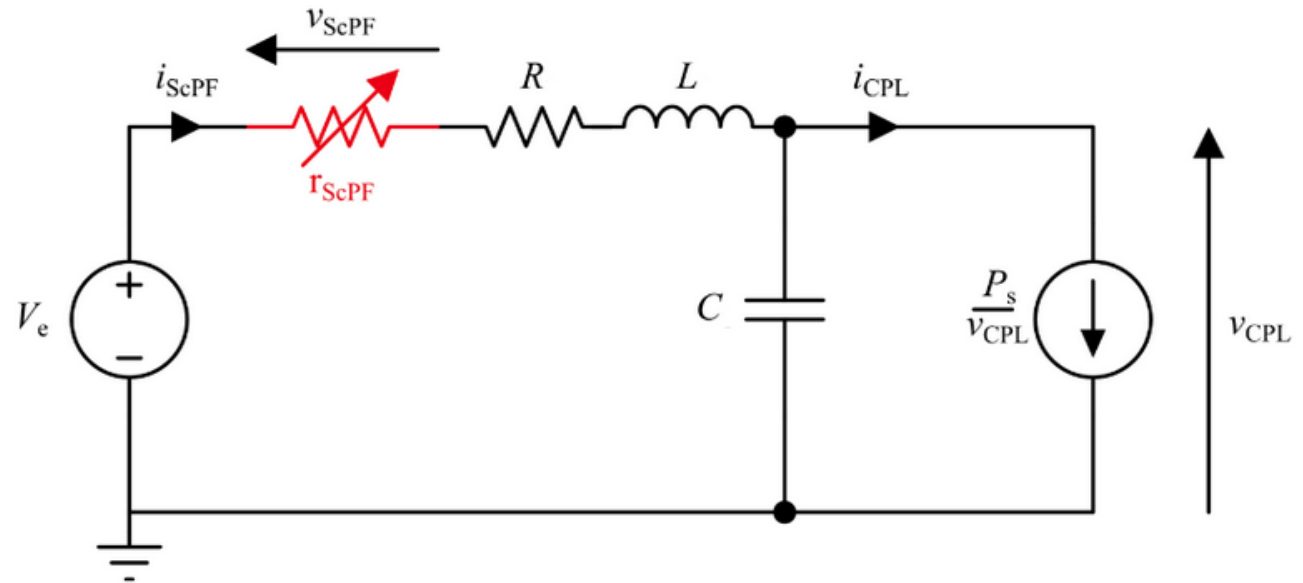


IV. Conclusion

Conclusion

5 parameters to optimize through 3 optimization problems different

- $Cost_{copper}$
- $Cost_{superconductor}$
- α : load power
- CAPEX
- OPEX





Thank you !



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